

GEOGRAPHY

PHYSICAL GEOGRAPHY



MODULE - 9

- EQUINOX
- VARIOUS ASPECTS OF REVOLUTION
- DAWN AND TWILIGHT
- PERIHELION AND APHELION

WHAT DO YOU UNDERSTAND BY EQUINOX?



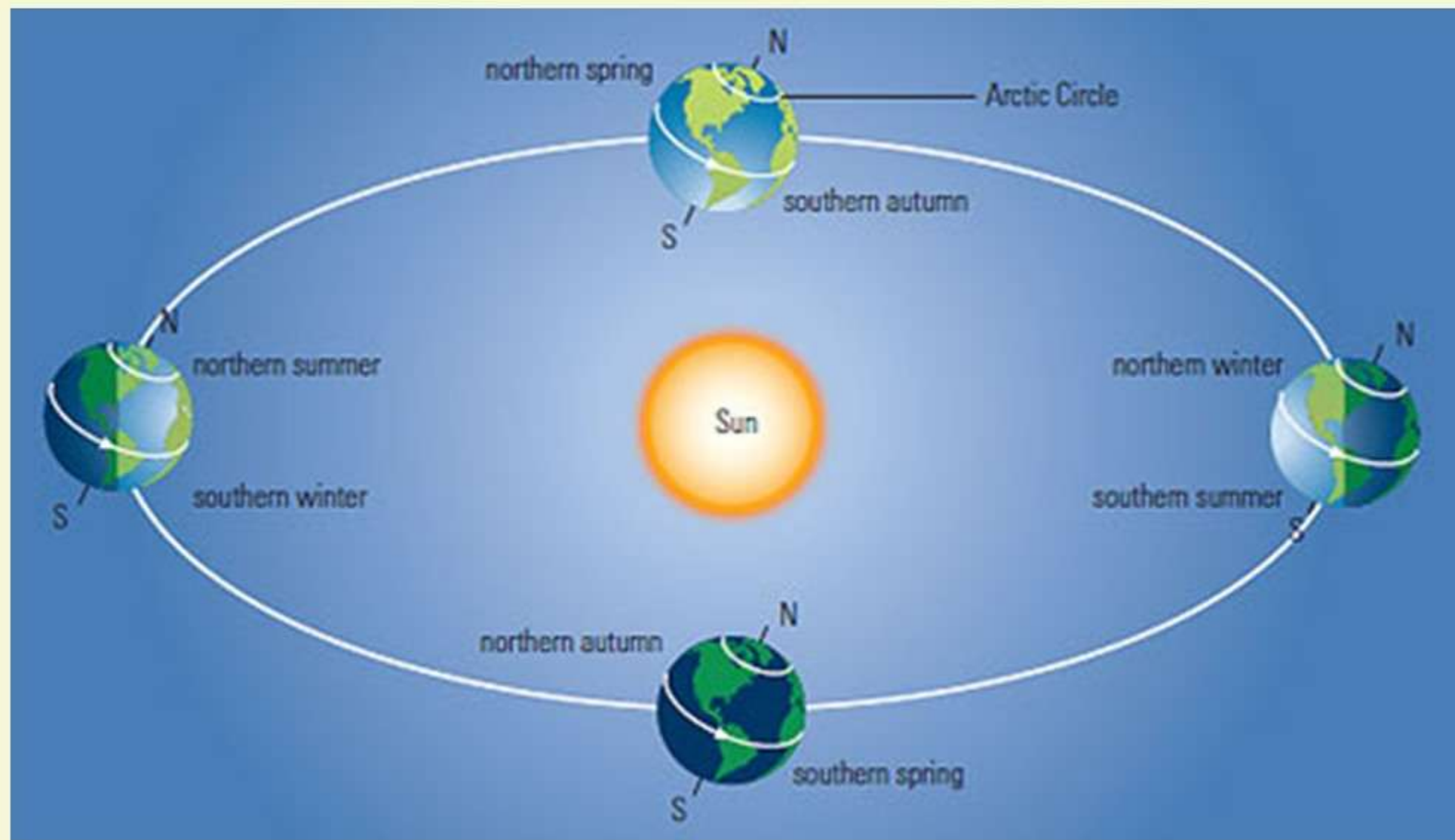
- 1 Equinox means equal nights.
- 2 We have seen Earth is actually tilted at an angle of around 23.4° towards the celestial pole.
- 3 On any other day of the year (other than Equinoxes), the Earth's axis also tilts a little away from or towards the Sun.
- 4 However, on the day of equinoxes twice a year, Earth's axis is not tilted towards or away from the Sun, but it is perpendicular to the Sun.

WHAT DO YOU UNDERSTAND BY EQUINOX?

- 5 However, the axial tilt of around 23.4° remains the same. But, on these two days, the tilt is perpendicular to the rays of the Sun.
- 6 On the day of two equinoxes, the tilt of Earth's axis is perpendicular to the Sun's rays.
- 7 Hence on the day of equinox, nights and days are nearly the same duration all over the world.

- 8 Equinox is derived from Latin and it means equal nights.
- 9 Most places enjoy more than 12 hrs of daylight on this day in reality.
- 10 This is because of the way, the sunrise and sunset are defined and also due to atmospheric refraction of sunlight.







HOWEVER, IT IS MORE THAN 12 HOURS OF DAY LIGHT ON THE DAY OF EQUINOXES

REASON NO: 1

- Sunrise is defined as the moment, the upper edge of the Sun's disk becomes visible above the horizon.
- At the same time, the sunset is referred to the moment, the Sun's upper edge disappears below the horizon.
- Because of the definition, it is more than 12 hrs.

REASON NO: 2

- The Earth's atmosphere also refracts sunlight, which is nothing but bending of the light.
- The Sun's upper edge will be visible from the Earth, several minutes before the edge actually reaches the horizon.
- The same phenomena happens at the sunset also.
- This causes every day on Earth to be at least 6 mins longer than it would have been without refraction.

Because of the refraction and the definition of sunrise and sunset, the day is actually longer than 12 hrs.

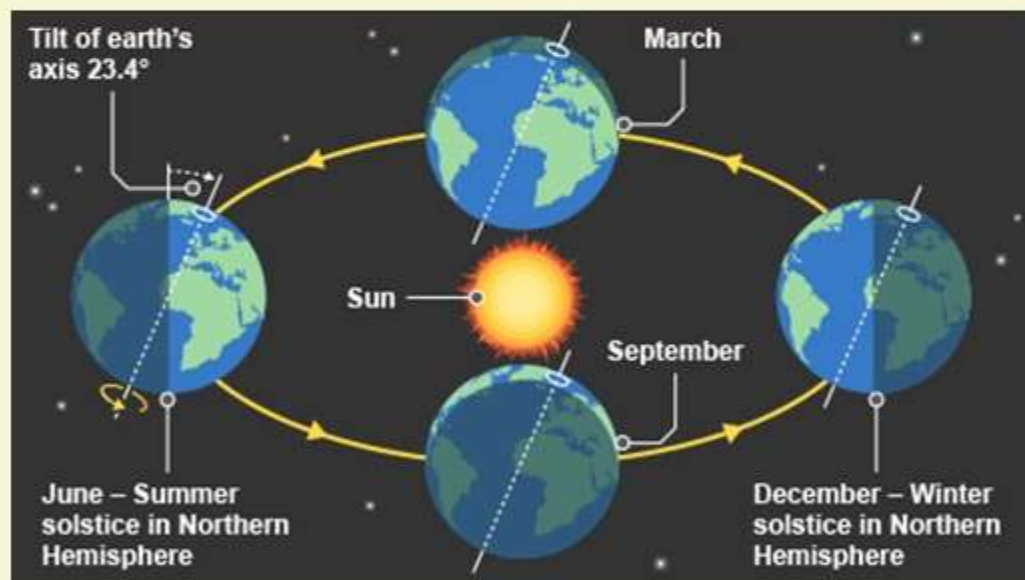




THE EARTH'S ROTATION AND REVOLUTION ... VARIOUS ASPECTS



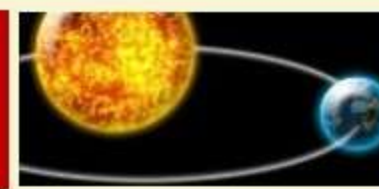
POINT NO: 1 VARYING LENGTHS OF DAY AND NIGHT



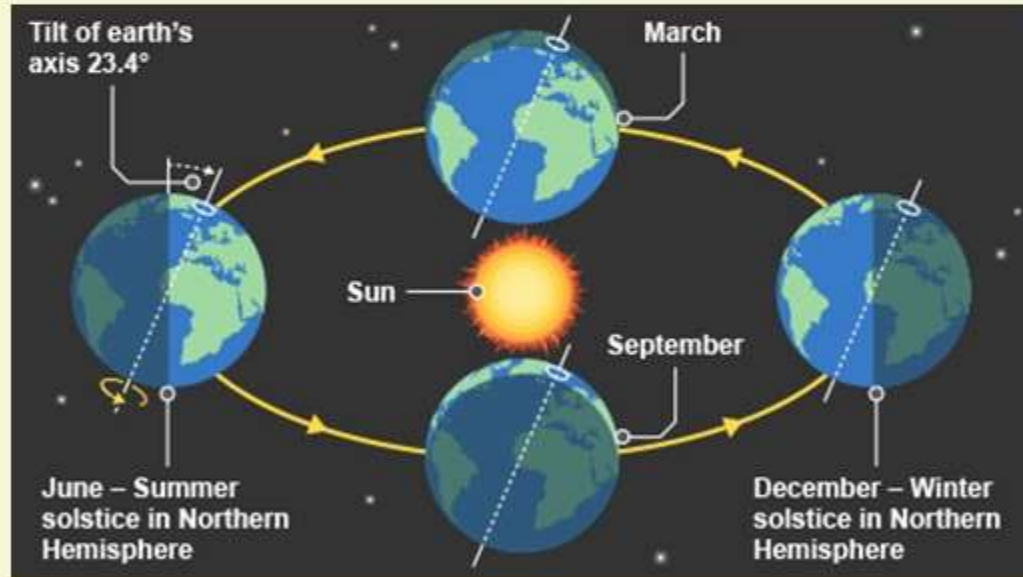
- 1 The axis of the Earth is inclined to the plane of the ecliptic at an angle of $66\frac{1}{2}^{\circ}$
- 2 It gives rise to different seasons and varying lengths of day and night.
- 3 Because of this tilt, at the Arctic circle ($66\frac{1}{2}^{\circ}\text{N}$), the Sun never rises and there is darkness for the whole day in mid-winter on 21st / 22nd December.
- 4 Beyond the Arctic circle, the number of days with complete darkness increases until we reach the North Pole.



THE EARTH'S ROTATION AND REVOLUTION ... VARIOUS ASPECTS



POINT NO: 1 VARYING LENGTHS OF DAY AND NIGHT



5 That means, there is darkness for almost half of the year and in summer, the conditions quite reverse and Sun never sets at the Arctic circle on 21st June and 24 hour period of continuous daylight and at the North pole there will be six months of continuous daylight.

6 Hence, the revolution of the Earth and its inclination to the plane of the ecliptic cause the variations in the lengths of day and night at different times of the year.

Tilt of earth's
axis 23.4°

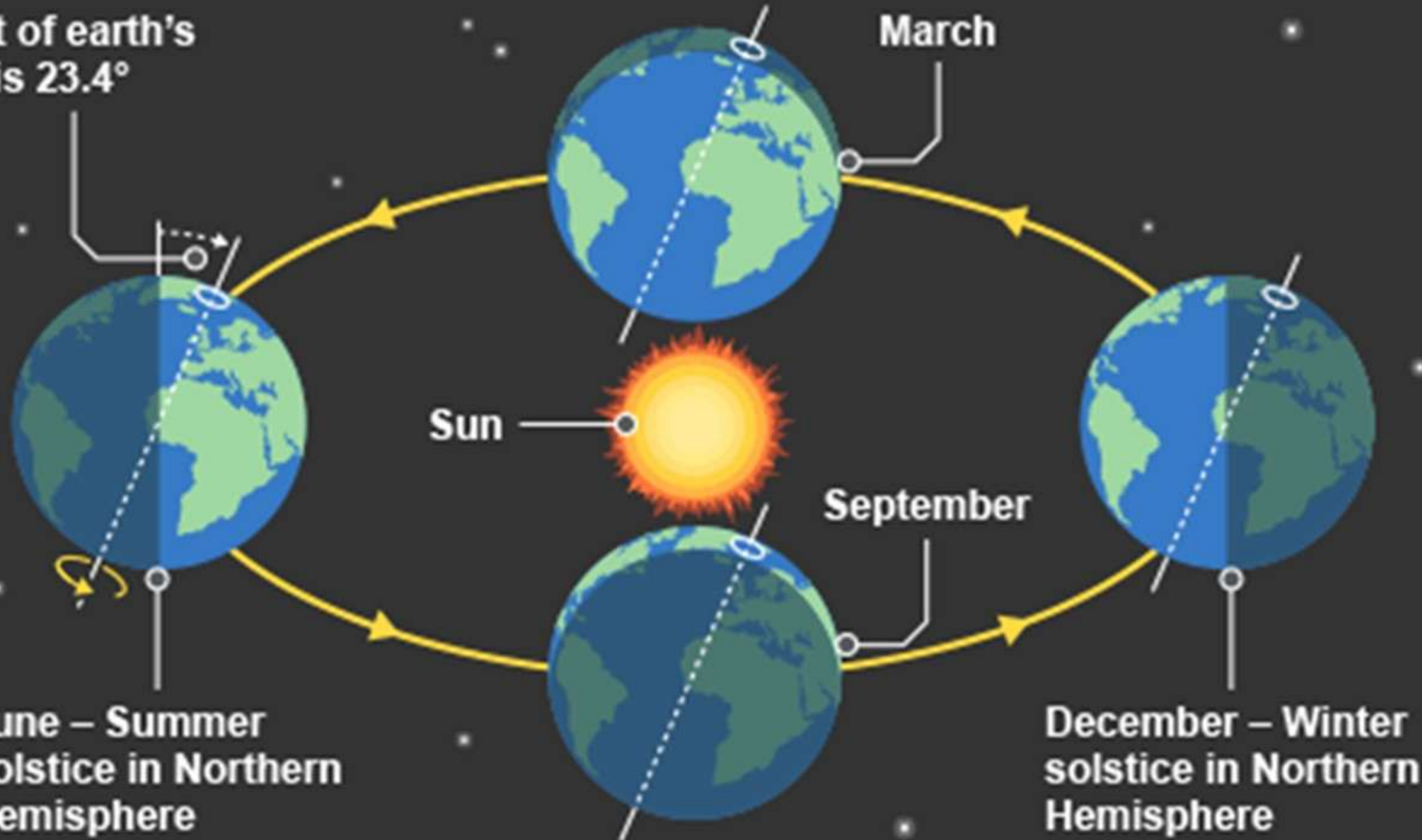
March

Sun

September

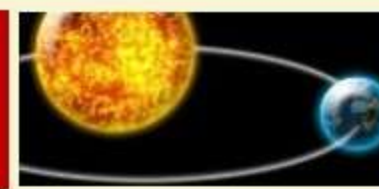
June – Summer
solstice in Northern
Hemisphere

December – Winter
solstice in Northern
Hemisphere

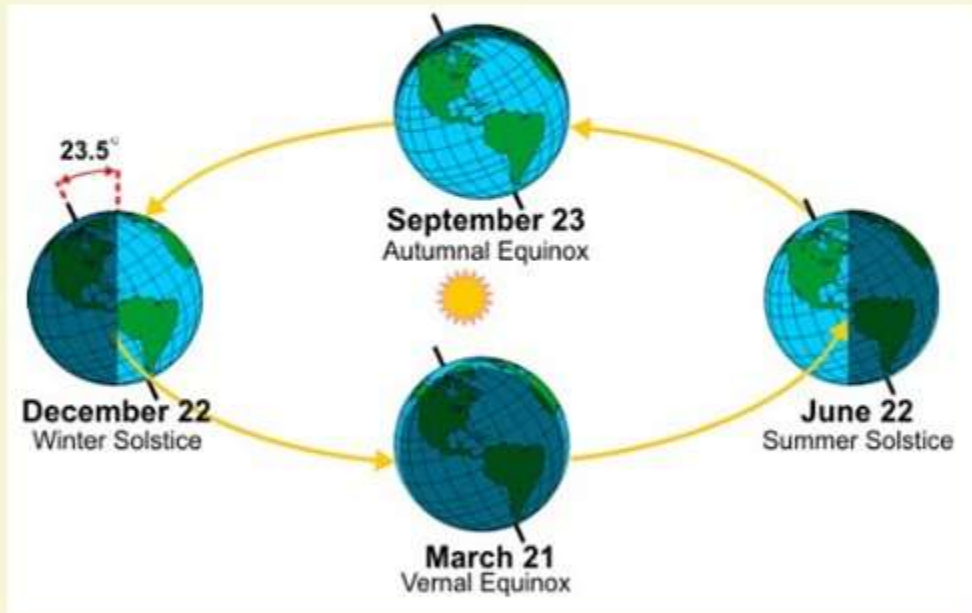




THE EARTH'S ROTATION AND REVOLUTION ... VARIOUS ASPECTS



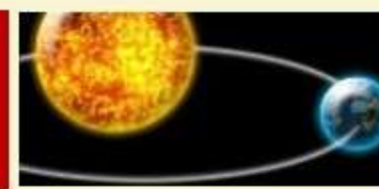
POINT NO: 2 THE ALTITUDE OF THE MID-DAY SUN



- 1 The Earth's revolution around the Sun with its axis inclined at $66\frac{1}{2}^\circ$ to the plane of the ecliptic changes the apparent altitude of the mid-day Sun.
- 2 The Sun is vertically overhead at the Equator on two days each year.
- 3 These are usually 21st March and 23rd September.
- 4 These are known as equinoxes, that means, equal nights and days.



THE EARTH'S ROTATION AND REVOLUTION ... VARIOUS ASPECTS



POINT NO: 2 THE ALTITUDE OF THE MID-DAY SUN

5 Then, after the March equinox, the Sun appears to move north and vertically overhead at the Tropic of Cancer on 21st June. This is known as **Summer Solstice in Northern hemisphere..**



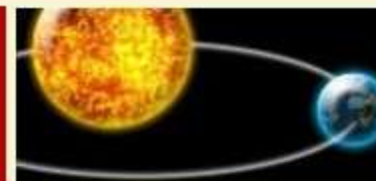
6 By 22nd Dec, the Sun will be overhead at the Tropic of Capricorn. This is known as **Winter Solstice in Northern hemisphere.**







THE EARTH'S ROTATION AND REVOLUTION ... VARIOUS ASPECTS



POINT NO: 2 THE ALTITUDE OF THE MID-DAY SUN

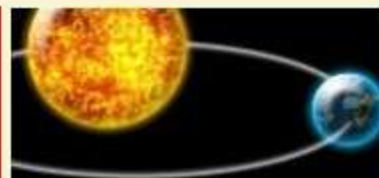
WHAT DOES THIS SIGNIFY?

1 The tropics marks the limits of the overhead Sun, beyond these the Sun is never overhead at any point of time. These are marked by seasonal changes like spring, summer, autumn and winter.

2 Beyond the Arctic Circle and the Antarctic Circle, it is always cold even during the short summer because of the reason, the Sun is never high in the sky.



THE EARTH'S ROTATION AND REVOLUTION ... VARIOUS ASPECTS



POINT NO: 3

SEASONAL CHANGES AND THEIR EFFECTS ON TEMPERATURE

1

- In summer, the Sun is higher in the sky than in winter.
- When the Sun is overhead, its rays fall almost vertically on the Earth, concentrating its heat on a small area.
- Temperature therefore rises and the summers are always warm.

2

- In winter, the rays are oblique.
- They come through the atmosphere less directly and much of their heat absorbed by the atmospheric impurities and water vapour.
- The Sun's rays fall faintly and spread over a great area, there is little heat and the temperatures remain low.

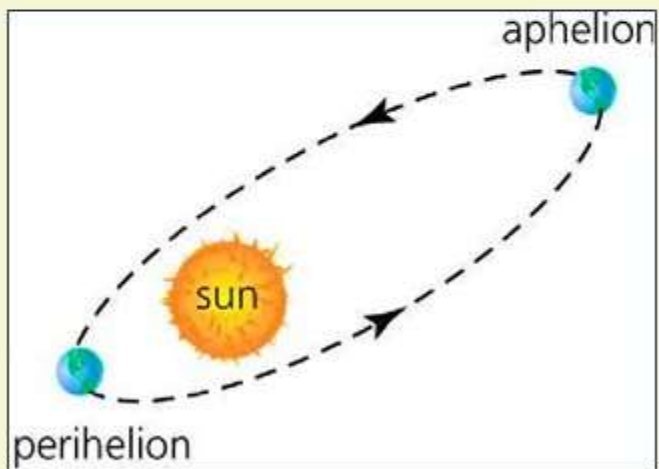
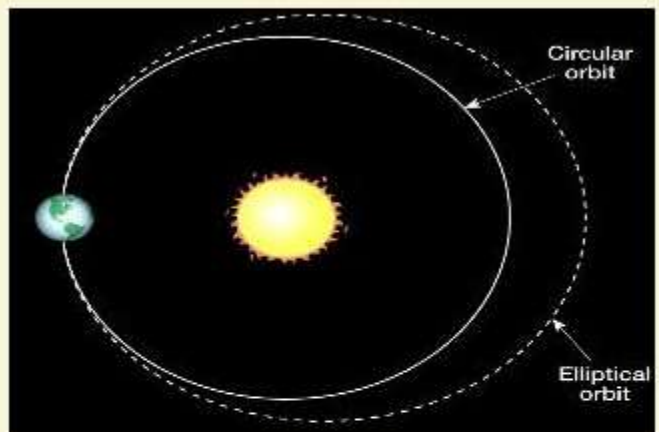


WHAT IS DAWN AND TWILIGHT?



- 1** The brief period between sunrise and full daylight is called '**dawn**' and that between sunset and complete darkness is termed '**twilight**'.
- 2** This is because of the reason, during the periods of dawn and twilight, the Earth receives diffused or refracted light from the Sun while it is still below the horizon.
- 3** Since, the Sun rises and sets in a vertical path at the equator, the period during which refracted light received is short.
- 4** But, in temperate latitudes, the Sun rises and sets in an oblique path and hence, the period of refracted light is longer.

REVOLUTION



PERIHELION AND APHELION

- The Earth is going around the sun in an **elliptical orbit** and throughout its orbit, the Earth is inclined in the same direction.
- The perihelion of any orbit of a celestial body about the Sun is the point where the body comes nearest to the Sun. It is the opposite of aphelion, which is the point in the orbit where the celestial body is farthest from the Sun.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

Amrita Naidu
D. Sunil Kumar
L. V. Krishna
D. Chaitanya

K. Srikanth
G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9